QAAP final research report

Methodology and findings from the completed run

The implemented pipeline classifies chest radiographs as COVID, Normal, Lung_Opacity or Viral Pneumonia. It combines a frozen lung segmenter, trainable quality-aware preprocessing and three convolutional classifiers. Their probabilities are combined by a logistic regression model trained on out-of-fold predictions. A separate calibration partition is used to fit confidence scaling before final testing.

On 2,111 held-out images, stacking achieved 95.17% accuracy and 95.52% macro-F1. Its advantage over soft voting was not statistically established. Single-scale attention outperformed the full multi-scale ConvNeXt, and QAAP gave no clear macro-F1 improvement. External transfer was weaker, and p95 end-to-end latency was 112.53 ms, above the 100 ms target.

1 Methodology

1 1 Data preparation and partitioning

The loader expects the four class folders of the COVID-19 Radiography Database [1]. Before splitting, it hashes decoded pixel arrays together with their shape and data type. Exact duplicate arrays are removed, and conflicting labels for the same array stop execution. This check does not detect every near-duplicate or establish independence between images from the same patient.

Three stratified splits allocate approximately 70% of the deduplicated images to training and 10% each to tuning, calibration and testing, with random seeds 42, 43 and 44. Integer rounding determines the final counts. Five stratified folds are formed within the 70% training partition. No patient grouping is implemented. The tuning partition is used for a segmentation audit and diagnostic model scores; the code does not select the best training epoch from these scores.

1 2 Intensity normalisation and lung cropping

Images are decoded at their native intensity depth and converted to grayscale when needed. Each image is scaled between its 0.5th and 99.5th intensity percentiles and clipped to [0, 1]. Unreadable, non-finite or constant images raise an error. This per-image transformation does not fit statistics across the evaluation cohort.

Lung localisation uses a frozen U-Net [2] released with GenSeg [3], whose repository identifies the checkpoint as trained with nine labelled JSRT examples. The segmentation input is quantised to 8 bits, histogram-equalised, gamma-corrected with gamma 0.5 and resized to 256 × 256. Predictions are thresholded at 0.5. The downloaded model and source files are hashed to record their provenance.

After resizing the mask to the original image, the code applies 5 × 5 morphological closing and retains at most two components, each covering at least 1% of the image. A valid crop requires two retained components and a combined area between 5% and 80%. Its bounding box receives an 8% margin on each axis. Invalid masks trigger a full-image fallback. Crops and masks are resized to 224 × 224; pixels outside the lung mask are not zeroed.

1 3 Quality features and learned preprocessing

Three features are measured on each cached crop: Laplacian variance for sharpness, intensity standard deviation for contrast, and the median absolute deviation of a residual for noise. The residual is the crop minus a 3 × 3 Gaussian-smoothed version. These are image statistics, not expert quality labels; anatomical texture and disease patterns may also affect them.

Each feature is divided by its 95th percentile in the relevant training partition, with a minimum divisor of 10^{-8}, then clipped to [0, 1]. Normalisers are fitted separately for each fold and once on the full training partition for the final models. Let s, c and n denote the resulting sharpness, contrast and noise values. The quality score is

$$q = w_s s + w_c c + w_n(1 - n), \quad w = softmax(a). \quad (1)$$

Here a contains three learned logits, so the weights are positive and sum to one. Equation (1) follows directly from LearnableQAAP. It implies that q lies in [0, 1]. Higher sharpness and contrast increase q, while higher estimated noise reduces it. The model does not learn an unrestricted image-quality function.

Write d = 1 − q for the degree of correction and let ℓ be the learned raw clip parameter. The local histogram clip limit is

$$L = 1 + softplus(\ell)d. \quad (2)$$

The soft histogram equaliser uses 64 bins in a 4 × 4 tile grid. A pixel contributes linearly to its two neighbouring intensity bins. Each normalised histogram is capped at L/64, and the excess is redistributed uniformly. The cumulative distribution is normalised after subtracting its first-bin value, with a 10^{-6} denominator floor. Spatial and intensity interpolation yield the equalised image E(x; L). This is the notebook's differentiable approximation to adaptive histogram equalisation.

$$y_1 = x + \alpha d[E(x; L) - x], \quad y_2 = y_1 + \beta d[B(y_1) - y_1]. \quad (3)$$

$$\tilde{x} = clip(y_2 + \gamma d[y_2 - G(y_2)], 0, 1). \quad (4)$$

The coefficients α, β and γ are sigmoids of learned scalars. B is a 5 × 5 bilateral filter with colour scale 0.1 and spatial scales 1.5. G is a 5 × 5 Gaussian filter with standard deviation 1.0. Equations (3) and (4) describe the actual order: equalisation, bilateral denoising, then unsharp masking. Bilateral filtering is an approximation used here in place of non-local means.

The logits start at zero, giving equal feature weights. The raw clip, equalisation, denoising and sharpening parameters start at 1, −1, −1 and −2. Each classifier learns its own preprocessing parameters through classification loss. The quality features remain fixed values measured before supervised augmentation; they are not recomputed after a flip or rotation. Preprocessing runs in float32 even when the classifier uses mixed precision.

1 4 Representation learning and classifier design

Self-supervised initialisation

ConvNeXt-small [4] starts from pretrained weights and receives one additional epoch of queue-based momentum contrast, following the query–key encoder design of MoCo [5]. This is a MoCo-style implementation with a memory queue; it should not be described as MoCo v3. Each fold trains its own encoder using only that fold's training images. A further encoder is trained on the full training partition for final ConvNeXt models.

Two independently augmented views are generated from each crop. Augmentation includes a random resized crop with area scale 0.75–1.0, horizontal flipping, rotation up to 10 degrees, brightness and contrast jitter of 0.15, and Gaussian blur. The SSL path uses RGB copies of an 8-bit crop and ImageNet normalisation. Learned QAAP is absent from this stage.

$$\theta_k \leftarrow 0.999\theta_k + 0.001\theta_q. \quad (5)$$

$$L_{SSL} = -log \frac{exp(u \cdot v^+ / \tau)}{exp(u \cdot v^+ / \tau) + \Sigma_j exp(u \cdot v_j^- / \tau)}, \quad \tau = 0.2. \quad (6)$$

In Equation (5), θq and θk are query and key encoder parameters. In Equation (6), u and v^+ are unit-normalised 128-dimensional projections of the paired views; v_j^- are the 65,536 queued negative vectors. The query encoder and its projector are trained by backpropagation. The key encoder follows the momentum update, and new keys enter the queue after the backward pass. The projection head is discarded before classification. These equations restate the implemented objective, rather than establish a new learning rule.

Convolutional models and transformer comparator

The main ConvNeXt model pools attention-refined feature maps from stages 1, 2 and 3 using zero-based indexing. Their 192, 384 and 768 channels form a 1,344-dimensional vector. Attention first weights channels using shared transformations of average- and max-pooled features, then weights spatial positions using a 7 × 7 convolution over channel-wise mean and maximum maps. This follows the channel-then-spatial pattern of CBAM [6], with a channel reduction factor of eight.

EfficientNetV2-S [7] uses the timm model tf_efficientnetv2_s.in21k_ft_in1k; DenseNet-201 [8] uses densenet201. Both apply attention to the final feature map before global average pooling. The transformer comparator is deit_small_patch16_224.fb_in1k [9]. It uses QAAP but has no added convolutional attention block or domain-specific SSL stage. The notebook does not perform teacher–student distillation.

Every classifier uses a 512-unit layer followed by LayerNorm, ReLU and dropout 0.4, then a 128-unit layer with LayerNorm, ReLU and dropout 0.3, and a four-class output layer. Grayscale inputs are replicated across three channels and normalised using the backbone's pretrained configuration. All supervised model parameters remain trainable.

1 5 Training stacking and calibration

Table 1 Training settings in the current notebook

Setting	Implemented value
Global batch size	32; incomplete SSL batches are dropped
Training schedule	1 SSL epoch; 5 supervised epochs per classifier
Optimiser	AdamW; weight decay 0.0001; fused mode enabled on CUDA
Initial learning rates	SSL 0.0003; supervised 0.0001
Schedule and precision	Epoch-wise cosine annealing; automatic mixed precision and gradient scaling
Supervised loss	Four-class cross-entropy with label smoothing 0.05
Supervised augmentation	Horizontal flip with probability 0.5; rotation uniformly within ±10°
Current resource defaults	Single GPU; zero loader workers; no pinned memory; cuDNN autotuning off

The fixed schedule saves the last epoch. There is no early stopping, class weighting or best-validation-checkpoint selection. A common epoch budget supports a comparison under limited training time, but does not establish that each architecture has converged. The configured random seed is 42; deterministic CUDA execution is not enforced.

Out-of-fold ensemble fitting

For each of five stratified folds, ConvNeXt, EfficientNetV2-S and DenseNet-201 are fitted on the other four folds. Fold-specific quality normalisers and SSL prevent those fitted components from using the held-out fold. The three four-class probability vectors are concatenated into 12 features for each held-out image. Once every training image has an out-of-fold prediction, logistic regression is fitted to these features and the training labels, with C = 1 and a maximum of 2,000 iterations.

At inference, separately trained final models produce the same 12 probability features. The fitted stacker maps them to four-class probabilities. An arithmetic mean of the three base probability vectors is also evaluated as soft voting. DeiT and the ConvNeXt ablation models are comparators; they do not enter either ensemble. Completing all scheduled jobs requires six SSL runs and 22 supervised model runs. All five folds and the final checkpoints were completed across staged sessions. The recovery notebook restored earlier folds before continuing the same experiment.

Probability calibration

A separate positive scalar temperature is fitted for every model and ensemble on the calibration partition by minimising negative log-likelihood, following temperature scaling [10]. The implementation searches log temperature between −3 and 3 and applies

$$p_{T,c} = \frac{exp(log(max(p_c, 10^{-12})) / T)}{\Sigma_j exp(log(max(p_j, 10^{-12})) / T)}. \quad (7)$$

Here the input is a class probability and T is the fitted temperature. The log probability equals the class logit minus a normalising term shared across classes. After division by T, that shared term cancels in softmax. Apart from clipping effects, this is logit temperature scaling. Positive scaling preserves the predicted class: accuracy and macro-F1 remain unchanged, while confidence metrics can change.

1 6 Evaluation and computational measurements

Internal test evaluation

The final report calculates accuracy, macro-F1, macro one-versus-rest ROC-AUC, Cohen's kappa and Matthews correlation coefficient. Confidence is assessed using negative log-likelihood, multiclass Brier score and expected calibration error across 15 equal-width confidence bins. Brier score is the per-image sum of squared class-probability errors, averaged over images; it is not divided by four. A confusion matrix and per-class classification report describe the stacking model's errors.

Uncertainty is estimated with 1,000 stratified bootstrap replicates, sampling with replacement within each test class and preserving its count. All models use the same draws. The 2.5th and 97.5th percentiles give intervals for accuracy, macro-F1 and AUC; paired macro-F1 differences compare stacking with each alternative. An exact two-sided binomial test on discordant correct/incorrect predictions implements the exact McNemar comparison, followed by Holm adjustment across the eight default comparisons. This test concerns correctness, not macro-F1. Image-level bootstrap intervals do not account for patient clustering or variation across training seeds.

External transfer and subgroup analysis

The Shenzhen and Montgomery datasets [12] are evaluated without retraining. The score is one minus the calibrated Normal probability, with abnormality predicted at a threshold of 0.5. TB is absent from the four training classes. Consequently, this evaluation tests transfer of a normal-versus-abnormal score to TB datasets; it does not evaluate a model trained specifically to diagnose TB.

Study identifiers join images to age, sex and optional manufacturer metadata. Duplicate metadata keys or unmatched image identifiers stop execution. Age is restricted to 0–120 years and grouped as below 40 or at least 40; missing values remain unknown. The code reports per-site sex, age and sex-by-age groups, with manufacturer groups only when real values are available. Each group includes its size and diagnosis counts. AUC is undefined when a group has only one diagnosis. AUC gaps are the maximum minus minimum among eligible known groups; subgroup uncertainty intervals are not implemented.

Localisation and efficiency

Grad-CAM [11] targets the predicted class at DenseNet's attention output. For each true class, the code selects the first available correct and incorrect prediction, giving at most eight examples. It records the fraction of heatmap intensity inside the predicted lung mask, returning an undefined value for invalid or fallback masks. These are selected case illustrations, not a random cohort-level localisation study.

The sparse equalisation backend samples the local lookup volume directly, avoiding the reference implementation's expanded batch × 64 × height × width tensor. At startup, synthetic forward/backward comparisons check output and parameter-gradient agreement and time both full QAAP paths. The faster path is selected, so the selected backend must be read from the benchmark record. This benchmark does not measure full training speed.

For end-to-end latency, ten warm-up calls precede batch-one inference on up to 50 test images. Synchronized timings include decoding, normalisation, segmentation, cropping, quality features, transfers, three classifiers, stacking and calibration. Mean, median and 95th-percentile latency are reported. The 100 ms p95 target is assessed against the measured total latency. Packed float32 crops are read one row at a time; checkpoints are saved every 100 batches, and resource measurements are logged every 50. Resource logs accompany the completed stages; their coverage and limits are reported with the measurements.

2 Ablation study

The completed component comparisons use the same 2,111 test images, the same training schedule and the same calibration procedure. All four ConvNeXt variants retain lung cropping, the custom classifier head and the additional SSL initialisation. Accuracy and macro-F1 are percentages; AUC and ECE are on a 0–1 scale.

Table 2 Measured ConvNeXt ablations

Variant	QAAP	Scales	Attention	Accuracy	Macro-F1	AUC
cn_vanilla*	Yes	Final	No	90.95	90.02	0.9831
cn_single	Yes	Final	Yes	93.08	93.29	0.9888
cn_noqaap	No	1, 2, 3	Yes	91.38	90.89	0.9819
cn full	Yes	1, 2, 3	Yes	91.33	90.97	0.9857

*The code name cn_vanilla removes added attention and multi-scale fusion. It still uses QAAP and SSL; its score is not a plain ImageNet-only ConvNeXt baseline.

Table 3 Paired macro F1 effects of the implemented components

Component	Contrast	Δ F1 pp	95% interval pp
Final-scale attention	cn_single minus cn_vanilla	+3.27	+1.94 to +4.62
Multi-scale fusion	cn minus cn_single	-2.31	-3.50 to -1.18
QAAP preprocessing	cn minus cn_noqaap	+0.08	-0.93 to +1.02

The paired intervals in Table 3 were calculated for this report from the saved test probabilities, using 1,000 stratified resamples and seed 42, as in the notebook. They were not present in the original ablation CSV. They are exploratory percentile intervals without multiplicity adjustment and describe test-image resampling for one trained run.

Interpretation of the component comparisons

Adding attention at the final scale increased macro-F1 by 3.27 percentage points, with an interval from 1.94 to 4.62. This supports attention within the single-scale configuration. Adding multi-scale fusion then reduced macro-F1 by 2.31 points, with an interval from −3.50 to −1.18. Under the five-epoch schedule, the larger fused representation did not improve the classifier. This contrast also changes the head input width from 768 to 1,344, so the result is not attributable to scale selection alone.

QAAP increased macro-F1 by only 0.08 points relative to cn_noqaap; its interval, −0.93 to 1.02, includes both a loss and a gain. Accuracy was 0.05 points lower with QAAP, although AUC increased from 0.9819 to 0.9857 and NLL fell from 0.2660 to 0.2564. The completed run therefore provides no clear macro-F1 benefit for QAAP in the full ConvNeXt configuration. The result does not isolate equalisation, denoising or sharpening.

2 1 Ensemble and remaining component comparisons

Table 4 Ensemble and preprocessing comparisons

Comparison	Measured finding	Inference
Stacking vs soft voting	Macro-F1 +0.48 pp; paired 95% interval −0.12 to +1.15 pp; adjusted p = 0.481	Small point-estimate advantage; superiority is not established.
Stacking calibration	ECE 0.0125 before and 0.0133 after; fitted T = 0.9866	Temperature scaling did not improve test ECE.
Sparse vs reference QAAP	27.15 vs 284.14 ms per forward/backward; 560.74 vs 765.87 MiB extra peak allocation	10.46-fold faster in this synthetic QAAP benchmark, with 26.78% lower extra peak allocation.

The ensemble p-value in Table 4 comes from the exact paired correctness test, with Holm adjustment across eight comparisons. It is not a test of macro-F1. The confidence interval concerns the paired F1 difference. The preprocessing benchmark uses batch 16 and does not measure full training speed or batch-one end-to-end latency.

Components not isolated in the completed run

The comparisons below remain outside the completed experiment. They are listed to define the scope of the measured ablations; no outcomes are assigned to them.

Table 5 Additional ablations needed for component attribution

Component	Required comparison	Status
Lung cropping	Retrain on full images with refitted quality scales, keeping the other settings fixed.	Not run
Additional SSL	Start the full ConvNeXt from ImageNet weights without the domain SSL epoch.	Not run
Quality weighting	Fix the three quality weights equally while training the remaining QAAP parameters.	Not run
Quality gating	Replace the adaptive correction strength and clip limit with a specified fixed policy.	Not run
Individual filters	Remove equalisation, bilateral denoising or sharpening separately, then retrain.	Not run
Multi-scale attention	Remove attention while retaining all three scales and the same head width.	Not run
Ensemble membership	Remove each base model in turn and refit stacking on the remaining OOF features.	Not run

There is one completed training realisation per configuration. Repeated seeds are needed to measure training variability. The paired bootstrap cannot establish that the same component ranking will hold after a longer schedule or across independently trained models. The earlier two-percentage-point acceptance check also remains unevaluated because a matching baseline run was not supplied to the completed notebook.

3 Results

3 1 Cohort and internal discrimination

The deduplicated cohort contained 21,106 unique decoded-image hashes. The archived manifest confirms the test identities in the prediction file. Training used 14,774 images; tuning used 2,110; calibration and testing each used 2,111. Table 6 gives the class counts after deduplication.

Table 6 Cohort composition by partition

Partition	COVID	Normal	Lung opacity	Viral pneumonia	Total
Train	2495	7134	4208	937	14774
Tune	356	1019	601	134	2110
Calibration	357	1019	601	134	2111
Test	357	1019	602	133	2111

Table 7 Internal test discrimination with 95% bootstrap intervals

Model	Accuracy %	Macro-F1 %	Macro AUC
ConvNeXt full	91.33 (90.19–92.61)	90.97 (89.54–92.35)	0.9857 (0.9821–0.9890)
EfficientNetV2-S	94.03 (92.99–95.07)	93.64 (92.43–94.80)	0.9925 (0.9901–0.9946)
DenseNet-201	94.22 (93.23–95.17)	94.31 (93.14–95.33)	0.9928 (0.9903–0.9950)
DeiT-small	94.50 (93.56–95.50)	94.55 (93.43–95.66)	0.9904 (0.9875–0.9932)
Soft voting	94.98 (94.08–95.97)	95.04 (94.03–96.09)	0.9934 (0.9910–0.9955)
Logistic stacking	95.17 (94.27–96.02)	95.52 (94.61–96.42)	0.9933 (0.9911–0.9954)

Stacking correctly classified 2,009 of 2,111 test images. Its accuracy was 95.17% (95% interval 94.27–96.02), macro-F1 was 95.52% (94.61–96.42), and macro one-versus-rest AUC was 0.9933 (0.9911–0.9954). Soft voting had slightly higher AUC, 0.9934, despite its lower accuracy and macro-F1. Among the six main comparisons, DeiT-small had the highest single-model accuracy and macro-F1; DenseNet-201 had the highest single-model AUC.

The three convolutional ensemble members had macro-F1 values of 90.97%, 93.64% and 94.31% for ConvNeXt, EfficientNetV2-S and DenseNet-201. Their combination improved the point estimates over each constituent. The stronger internal score for the ensemble should be interpreted alongside the paired comparisons and the external transfer results, rather than treated as evidence that all components contributed equally.

Consistency checks

Accuracy, macro-F1, ECE, NLL and Brier score were recalculated from saved test probabilities and agreed with the exported values within numerical precision. The same predictions reproduce the confusion matrix and per-class results. The completed notebook retains the original top-level model and training function definitions; its additional cells organise recovery, staged training and report backups.

3 2 Paired comparisons and confidence quality

Table 8 Stacking compared with each alternative

Alternative	Δ F1 pp	95% interval pp	Holm p
ConvNeXt full	+4.54	+3.39 to +5.77	8.81e-16
EfficientNetV2-S	+1.87	+0.87 to +2.91	0.0189
DenseNet-201	+1.21	+0.35 to +2.07	0.0181
DeiT-small	+0.96	-0.04 to +2.01	0.26
ConvNeXt vanilla*	+5.50	+4.24 to +6.98	6.6e-15
ConvNeXt single scale	+2.23	+1.23 to +3.21	3.17e-05
ConvNeXt without QAAP	+4.63	+3.42 to +5.82	9.96e-13
Soft voting	+0.48	-0.12 to +1.15	0.481

Stacking improved on the three convolutional members by 4.54, 1.87 and 1.21 macro-F1 points. Each paired F1 interval was positive, and the adjusted correctness-test p-values were below 0.05. Its estimated gains over DeiT-small and soft voting were 0.96 and 0.48 points, but their F1 intervals crossed zero and adjusted p-values were 0.260 and 0.481. Those comparisons do not establish superiority. For stacking versus soft voting, 11 images were correct only under stacking and seven only under soft voting.

Table 9 Calibrated internal confidence and agreement metrics

Model	ECE	NLL	Brier	Kappa	MCC
ConvNeXt full	0.0230	0.2564	0.1302	0.8679	0.8681
EfficientNetV2-S	0.0150	0.2019	0.0967	0.9093	0.9096
DenseNet-201	0.0164	0.1873	0.0917	0.9119	0.9119
DeiT-small	0.0130	0.1908	0.0918	0.9156	0.9157
Soft voting	0.0115	0.1604	0.0800	0.9233	0.9234
Logistic stacking	0.0133	0.1593	0.0799	0.9260	0.9260

Stacking had the lowest NLL and Brier point estimates, 0.1593 and 0.0799. Soft voting had lower ECE, 0.0115 versus 0.0133. These metrics measure different aspects of probability quality, so the rankings need not coincide. Kappa and MCC were both approximately 0.926 for stacking. The notebook does not provide uncertainty intervals for these confidence or agreement metrics.

All intervals and tests concern the held-out images from this run. They do not capture variability across training seeds, and their independence assumption remains limited by image-level splitting without verified patient identifiers. The adjusted p-values refer specifically to the eight stacking comparisons; they do not supply multiplicity control for the additional component intervals in Table 3.

3 3 Class level errors

Table 10 Stacking precision recall and F1 by class

Class	Test n	Precision %	Recall %	F1 %
COVID	357	96.30	94.68	95.48
Normal	1019	95.69	95.98	95.84
Lung opacity	602	93.21	93.52	93.37
Viral pneumonia	133	97.01	97.74	97.38

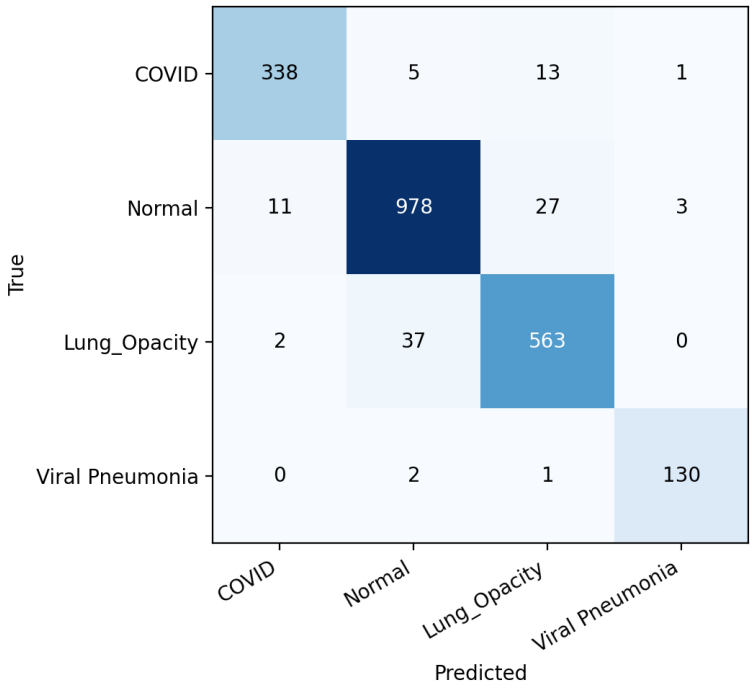


Figure 1 Internal test confusion matrix for calibrated stacking. Rows are true labels; columns are predicted labels. Source: completed run.

Normal and Lung opacity accounted for 64 of the 102 errors: 37 Lung opacity images were predicted Normal, and 27 Normal images were predicted Lung opacity. This was the largest confusion pair. COVID recall was 94.68%, with 13 COVID images assigned to Lung opacity and five to Normal. Viral pneumonia had the highest F1, 97.38%, but only 133 test images represented that class. The smaller support should be retained when comparing class-specific point estimates.

3 4 Temperature scaling

Table 11 Stacking confidence before and after temperature scaling

Metric	Before	After
ECE	0.0125	0.0133
NLL	0.1591	0.1593
Brier score	0.0798	0.0799

The fitted stacking temperature was 0.9866, close to one. Test ECE increased from 0.0125 to 0.0133 after calibration. NLL and Brier score also increased slightly. Temperature scaling was fitted to calibration NLL, so an improvement on the test cohort was never guaranteed. Accuracy and macro-F1 were unchanged because positive scalar scaling preserves the class with the highest probability.

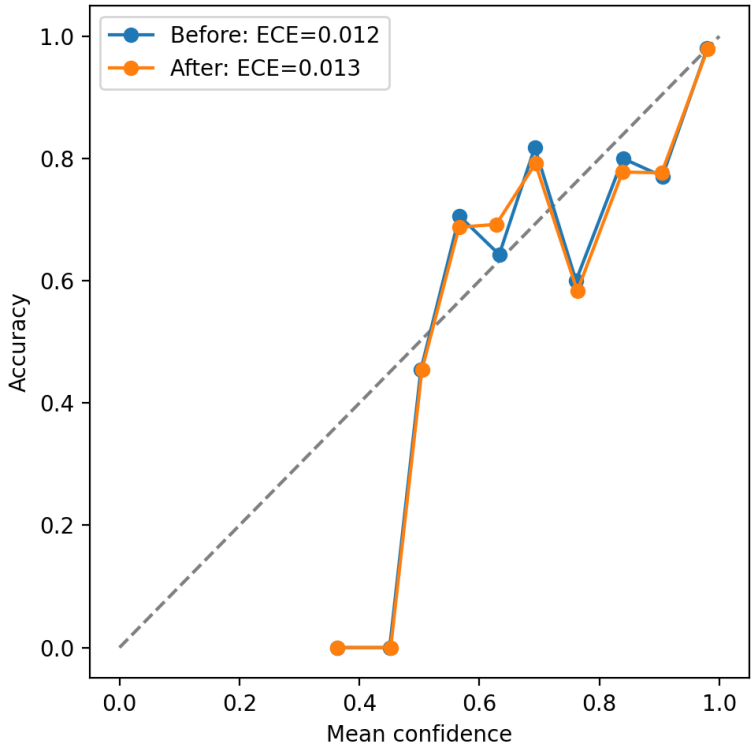


Figure 2 Stacking reliability before and after temperature scaling. The diagonal indicates equality between mean confidence and empirical accuracy. The plotted ECE labels are rounded to three decimals. Source: completed run.

The before-calibration values in Table 11 were reconstructed for this report by raising the saved calibrated probabilities to the fitted temperature and normalising each row. This inverts the notebook's temperature transformation for the stacking probabilities, whose values are above the clipping floor. The reconstruction does not involve refitting. The exported reliability plot independently shows the same ECE direction. Sparse confidence bins can produce unstable individual points; their positions should not be read as uniform error across the test cohort.

3 5 External normal versus abnormal transfer

External evaluation comprised 800 radiographs: 662 from Shenzhen and 138 from Montgomery, with 406 Normal and 394 TB-labelled images in total. Models were applied without external retraining. The score was one minus the calibrated Normal probability. The reported sensitivity and specificity use a fixed threshold of 0.5 and concern abnormality transfer, since TB was absent from the training classes.

Table 12 Pooled external transfer results

Model	Accuracy %	Macro-F1 %	AUC	Sensitivity %	Specificity %
ConvNeXt full	77.25	77.19	0.8311	73.35	81.03
EfficientNetV2-S	76.75	76.75	0.8370	76.65	76.85
DenseNet-201	67.75	67.63	0.7628	74.87	60.84
DeiT-small	65.50	62.62	0.7702	38.32	91.87
ConvNeXt vanilla*	70.62	69.39	0.7942	51.27	89.41
ConvNeXt single scale	78.50	78.42	0.8591	73.35	83.50
ConvNeXt without QAAP	76.00	75.68	0.8326	65.48	86.21
Soft voting	77.75	77.73	0.8518	75.89	79.56
Logistic stacking	75.88	75.29	0.8470	61.42	89.90

Table 13 Stacking transfer by site

Site	Normal / TB	Accuracy %	AUC	Sensitivity %	Specificity %
Montgomery	80 / 58	77.54	0.8203	56.90	92.50
Shenzhen	326 / 336	75.53	0.8502	62.20	89.26

Stacking achieved 75.88% pooled accuracy, 75.29% macro-F1 and AUC 0.8470. It identified 242 of 394 TB-labelled images as abnormal and correctly retained 365 of 406 Normal images, giving sensitivity 61.42% and specificity 89.90%. Its high internal score did not translate into the strongest external result. Single-scale ConvNeXt had the highest pooled accuracy, macro-F1 and AUC among the evaluated models; soft voting also exceeded stacking on those point estimates.

The internal and external tasks have different labels and score definitions, so the difference between their F1 values is not a direct estimate of a same-task generalisation loss. The external ranking nevertheless shows that selecting the strongest internal model does not establish the strongest abnormality transfer. No external paired tests or confidence intervals were exported; differences in Table 12 remain descriptive.

3 6 Demographic findings

Age was available for all 800 external records and sex for 799. Table 14 reports stacking at the fixed abnormality threshold. Group sizes include both diagnoses; the Normal/TB counts show the denominators behind sensitivity and specificity.

Table 14 External stacking results by sex and age

Site and group	Normal / TB	AUC	Sensitivity %	Specificity %
Montgomery Female	53 / 21	0.8257	71.43	90.57
Montgomery Male	26 / 37	0.8493	48.65	96.15
Montgomery 40+	33 / 37	0.8247	70.27	84.85
Montgomery <40	47 / 21	0.7741	33.33	97.87
Shenzhen Female	101 / 101	0.8294	54.46	92.08
Shenzhen Male	225 / 235	0.8586	65.53	88.00
Shenzhen 40+	85 / 116	0.8861	74.14	94.12
Shenzhen <40	241 / 220	0.8290	55.91	87.55

In Montgomery, sensitivity was 71.43% for Female records and 48.65% for Male records, a 22.78-point difference. Shenzhen showed the opposite direction: 54.46% for Female and 65.53% for Male records. A pooled sex comparison would conceal this site dependence. Age differences were also visible: sensitivity below 40 was 33.33% in Montgomery and 55.91% in Shenzhen, compared with 70.27% and 74.14% in the respective older groups.

Table 15 AUC gaps across known demographic groups

Site	Sex gap	Age gap	Sex by age gap
Montgomery	0.0236	0.0507	0.1238
Shenzhen	0.0291	0.0571	0.0869

The lowest intersectional sensitivity was 26.67% among Montgomery Male records below 40, based on 15 TB-labelled images (four predicted abnormal). Montgomery Female records below 40 included only six TB-labelled images. Such small denominators can produce large changes from a few predictions. The subgroup analysis has no confidence intervals, adjustment for case mix or demographic hypothesis tests; it identifies patterns for further study rather than an established causal demographic effect.

Manufacturer information was unavailable for every external image: zero of 800 records had known values. No manufacturer disparity estimate can therefore be reported. The one Montgomery record with unknown sex had a Normal label and was excluded from known-sex comparisons; its AUC was undefined.

3 7 Segmentation and localisation

Table 16 Frozen U Net segmentation audit on tuning images

Class	Audited n	Mean Dice	Minimum Dice
COVID	21	0.8819	0.5239
Normal	52	0.9624	0.8348
Lung opacity	25	0.8058	0.3892
Viral pneumonia	2	0.5495	0.2427

Across 100 audited tuning images, mean Dice was 0.8981 and median Dice was 0.9554. Normal images had the highest mean Dice, 0.9624. Lung opacity images had mean Dice 0.8058 and a minimum of 0.3892. The Viral pneumonia mean was 0.5495, but it was based on only two images and cannot establish the class-wide segmentation rate. The spread indicates that the frozen segmenter sometimes failed despite a high overall median.

Table 17 DenseNet Grad CAM lung heat fractions

True class	Prediction	Correct	Heat in mask %
COVID	COVID	Yes	19.24
COVID	Normal	No	49.80
Normal	Normal	Yes	46.67
Normal	COVID	No	22.78
Lung opacity	Lung opacity	Yes	34.82
Lung opacity	Normal	No	25.35
Viral pneumonia	Viral pneumonia	Yes	26.76
Viral pneumonia	Normal	No	8.57

None of the eight selected examples used the crop fallback. Lung heat fractions ranged from 8.57% to 49.80%, with a descriptive mean of 29.25%. This does not support a claim that most Grad-CAM mass lay inside the lung masks. It also does not prove that the model ignored lung disease: the fraction depends on mask area, mask accuracy and the coarse spatial resolution of the heatmap. No lesion annotations or mask-area-normalised baseline were used.

The examples were the first available correct and incorrect DenseNet prediction for each class. They were not randomly sampled, and the heatmaps explain the DenseNet component rather than the stacked ensemble. Selected visual examples are shown on the following page to retain the distinction between a class prediction and its spatial explanation.

3 8 Selected Grad CAM examples

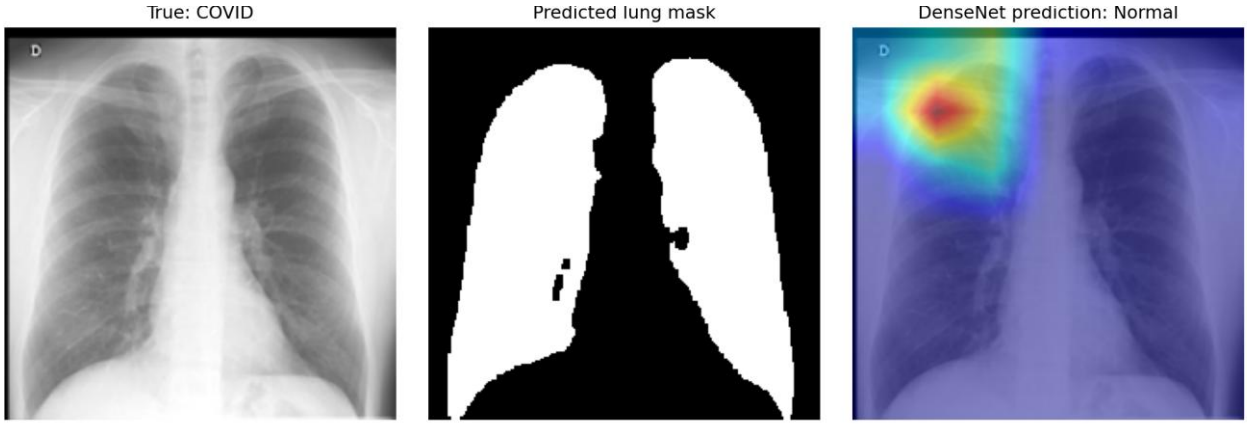


Figure 3 COVID image predicted Normal by DenseNet. The heat fraction inside the predicted lung mask was 49.80%. The left panel is the crop, the middle is the predicted mask, and the right is the Grad-CAM overlay. Source: completed run.

The COVID error assigns concentrated heat to the upper lung region on the left of the displayed image. The numerical overlap is relatively high among these examples, yet the class prediction is incorrect. Mask overlap alone therefore cannot validate the diagnostic interpretation.

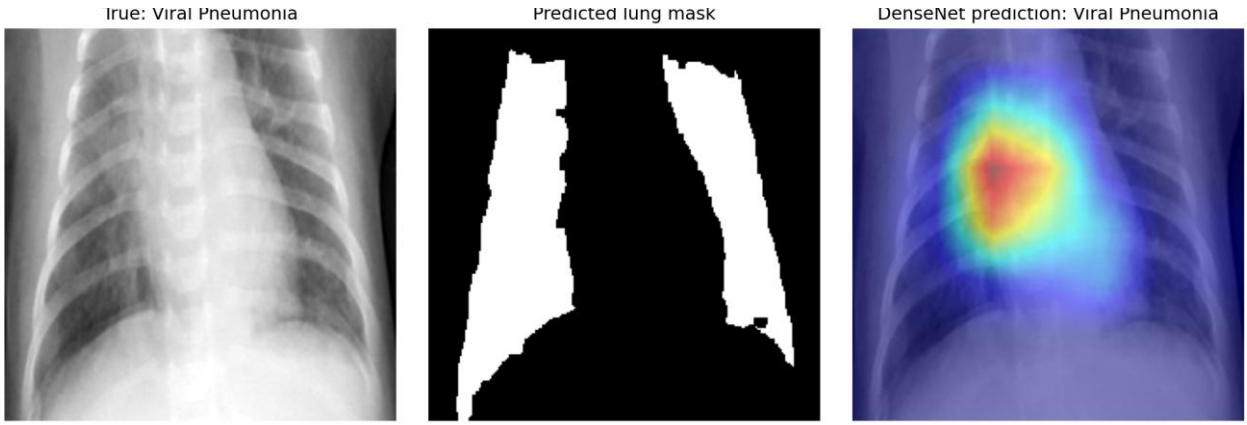


Figure 4 Viral pneumonia image correctly classified by DenseNet. Only 26.76% of the heatmap mass lies inside the predicted mask. Source: completed run.

The second image is correctly classified despite limited overlap between the highlighted central region and the predicted lung mask. The mask excludes part of the central thorax, so the overlap score partly reflects segmentation geometry. Both examples warrant review alongside segmentation quality rather than a single localisation threshold.

3 9 Computational performance and recorded resources

Table 18 Synthetic full QAAP forward and backward benchmark

Backend	Batch	Time ms	Extra peak MiB
Reference	16	284.14	765.87
Sparse	16	27.15	560.74

The sparse backend was selected on a Tesla T4 after the notebook's output and gradient agreement checks passed. It reduced synthetic QAAP forward/backward time by 90.44%, corresponding to a 10.46-fold speed ratio, and reduced extra peak GPU allocation by 26.78%. These numbers apply to batch 16 in this benchmark. They do not measure total training acceleration or peak whole-model memory.

Table 19 Batch one latency on 50 images after ten warm up calls

Stage	Mean ms	Median ms	p95 ms
Decode and normalise	6.11	5.09	9.33
Lung segmentation	12.86	12.83	13.24
ROI crop	2.34	2.32	2.68
Quality features	1.91	1.90	2.17
ConvNeXt classification	17.78	17.54	19.94
EfficientNet classification	23.58	23.43	25.16
DenseNet classification	33.43	33.28	34.94
Stacking and calibration	0.44	0.44	0.50
End to end total	107.62	106.69	112.53

End-to-end mean, median and p95 latency were 107.62, 106.69 and 112.53 ms. The p95 exceeded the 100 ms target by 12.53 ms (12.53%). DenseNet classification was the largest named stage at 33.43 ms mean. The three QAAP preprocessing steps added approximately 8.76 ms mean in total; transfers and other recorded stages are included in the total. Stage p95 values are not additive.

The recorded environment used PyTorch 2.11.0 with CUDA 12.8, timm 1.0.15 and Kornia 0.7.4. Two Tesla T4 devices were visible, while the final execution policy specified single-GPU models, zero loader workers, no pinned memory and cuDNN autotuning off. Latency was measured on a single GPU with all three classifiers available in memory.

The merged logs contain 916 resource samples and 116 epoch-timing entries. Their maximum sampled process RSS was 20.44 GiB and maximum cgroup usage was 27.46 GiB against a 30 GiB limit. The last logged process RSS was 3.58 GiB. Logged training segments sum to 3.61 hours, but exclude preprocessing, evaluation, session overhead and earlier unsaved or interrupted work; the first SSL entry covers only the resumed remainder. These records cannot establish a complete wall-clock runtime or a continuous memory peak.

4 Inferences and research limits

Ensembling improved the internal result

The clearest internal gain was the combination of the convolutional models. Stacking exceeded each constituent in macro-F1, and the paired intervals were positive. Its advantage over soft voting was less certain: a 0.48-point gain with an interval crossing zero and only four additional correct predictions. For this run, soft voting remains a close alternative. A preference for stacking should therefore depend on the small measured gain and its added fitting step, rather than a claim of established superiority over averaging.

The full architecture was not the strongest ConvNeXt variant

The ablations support final-scale attention, but they do not support the assumed benefit of multi-scale fusion. Single-scale attention achieved 93.29% macro-F1 against 90.97% for the full model, and it also had the strongest pooled external transfer. This is evidence to revisit the fused representation under the present schedule. QAAP's F1 effect was too small and uncertain to establish a benefit. Claims that all preprocessing stages or all architectural additions improved performance would contradict the completed comparisons.

Internal discrimination did not settle external transfer

Stacking's external sensitivity was 61.42% at threshold 0.5, despite high internal discrimination. Soft voting and single-scale ConvNeXt had stronger pooled external point estimates. The external task uses TB-labelled images and a transferred abnormality score, so it cannot validate TB-specific diagnosis or calibrated external risk. Site and demographic variation further limit a general screening claim. Any threshold adaptation should be fitted on separate external development data and assessed on a new evaluation cohort.

Confidence and localisation claims need narrower wording

Temperature scaling produced little change in stacking probabilities and slightly worsened test ECE. The final report can state the calibrated scores, but cannot claim that calibration improved every confidence measure. The segmentation audit also exposed poor masks in some images. Grad-CAM overlap was below 50% in all eight selected cases, with no expert lesion annotations. These results justify inspection of failure cases; they do not validate ensemble localisation or establish the absence of shortcuts.

Remaining uncertainty concerns patients and repeated training

Exact pixel deduplication and out-of-fold fitting reduce specific leakage paths. They do not establish patient-level independence or remove near-duplicates and acquisition cues. The bootstrap resamples images from one completed run, while subgroup comparisons and latency use limited samples. A fixed five-epoch supervised schedule gives a practical comparison under one budget, not convergence evidence for every backbone. The recorded benchmark supports a faster, smaller QAAP computation, but full-pipeline p95 remained above the target and a same-cohort earlier-run acceptance comparison was not completed.

References and implementation provenance

The numbered references identify borrowed architectures, learning procedures and the public segmentation checkpoint. Equations (1)–(4) are a mathematical restatement of the notebook's QAAP code; Equations (5)–(6) express its MoCo-style objective; Equation (7) expresses its temperature transformation. The argument about unchanged class predictions follows algebraically from positive scalar scaling. No claim of novelty is assigned to these established components.

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